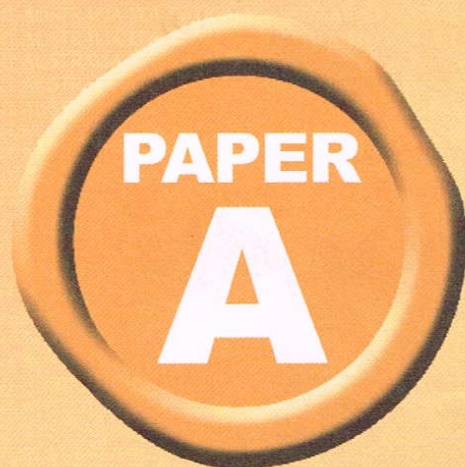




ICAS

INTERNATIONAL COMPETITIONS AND
ASSESSMENTS FOR SCHOOLS
MATHEMATICS
2009

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

**PLEASE SEE BACK COVER FOR A LIST
OF THE YEAR LEVELS THAT SHOULD
SIT THIS PAPER**

1. The picture shows a herd of dinosaurs.



How many dinosaurs are in the herd?

- (A) 6
- (B) 7
- (C) 8
- (D) 9

2. Rema marked the dates of her basketball practice on the calendar.

March

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

 = basketball practice

What was the date of her fourth basketball practice in March?

- (A) 4 March
- (B) 6 March
- (C) 14 March
- (D) 18 March

3. Which shape has the most edges?



(A)



(B)



(C)



(D)

4. Ann had 12 tomatoes.



She ate 5 of them.

How many tomatoes did she have left?

- (A) 3
- (B) 7
- (C) 8
- (D) 17

5. The picture shows four pieces of string.

Which piece of string is longest?



(A)



(B)

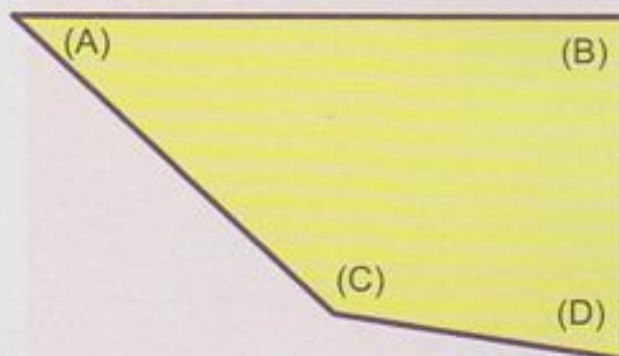


(C)

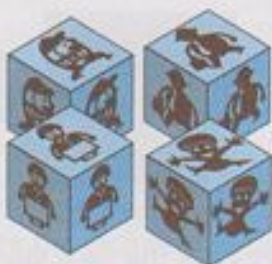


(D)

6. Which is the smallest angle in this shape?



7. Omar had four cubes.



He moved one of the cubes.



Which cube did he move?



(A)



(B)



(C)



(D)

8. Boris made a pattern of shapes. The picture shows the start of his pattern.



Which shape comes next in Boris's pattern?



(A)



(B)

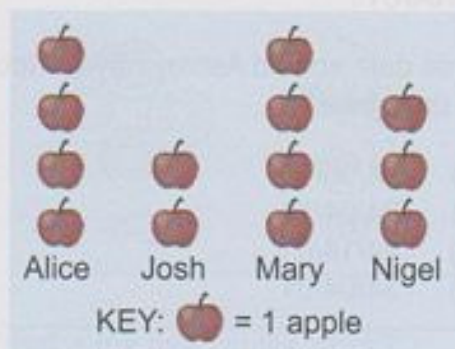


(C)



(D)

9. Four children were given some apples, as shown in the diagram.

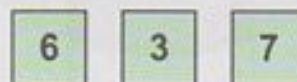


Nigel ate Mary's apples and his own apples.

How many apples did he eat?

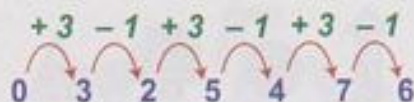
- (A) 1
(B) 3
(C) 4
(D) 7

10. What is the biggest number that you can make using the following digits?



- (A) 763
(B) 736
(C) 673
(D) 367

11. The picture shows how Simon made a number pattern.



What number comes next in Simon's pattern?

- (A) 3
- (B) 5
- (C) 7
- (D) 9

12. Ashley wrote the date in her diary as '24/04/01'.

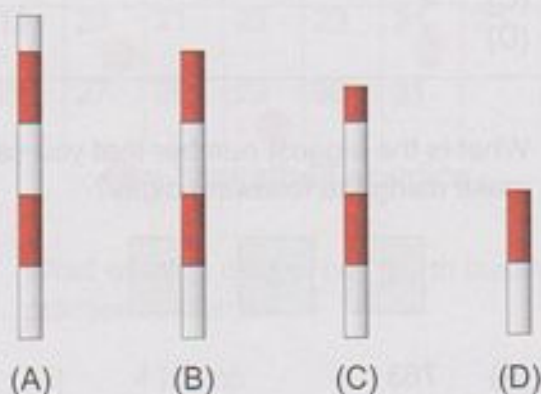
What date should Ashley have written 10 days later?

- (A) 04/05/01
- (B) 24/04/11
- (C) 24/14/01
- (D) 34/04/01

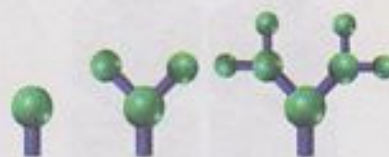
13. A stick is painted with equal strips of red and white.



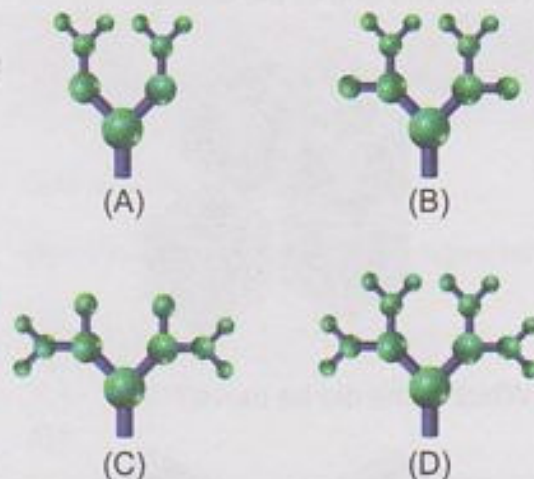
Which picture shows half of this stick?



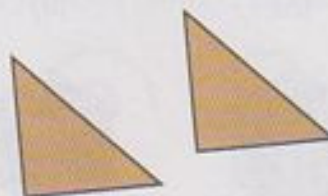
14. Sam uses sticks and balls to make a pattern.



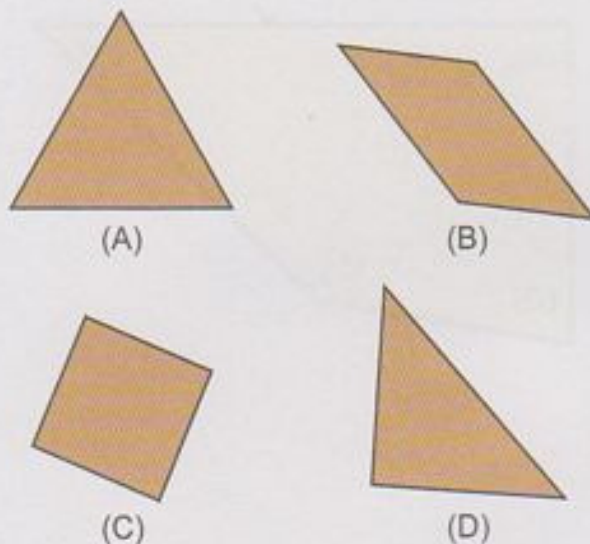
Which picture is the next step in Sam's pattern?



15. Here are two identical shapes.



Which shape **cannot** be made by putting the shapes together along an edge?



16. Bella holds this picture in front of a mirror.



How does the picture appear in the mirror?



(A)



(B)



(C)



(D)

17. Sasi has these cards.



She picks one card without looking.

Which card is she most likely to pick?



(A)



(B)

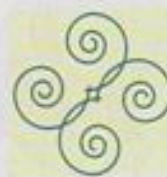


(C)



(D)

18. Celia drew this pattern. It has two lines of symmetry.



Which of these patterns also has two lines of symmetry?



(A)



(B)



(C)



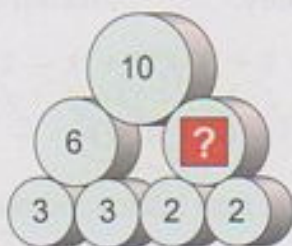
(D)

19. What is the time on the clock?



- (A) 1:04
(B) 1:20
(C) 4:01
(D) 4:05

20. Here is a number pattern.



What is the missing number?

- (A) 7
- (B) 6
- (C) 5
- (D) 4

22. Students at Happyville School can earn Bronze, Silver and Gold awards.

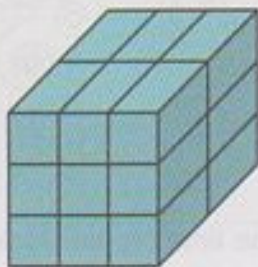
Students with 6 Bronze awards receive a Silver award.

Students with 4 Silver awards receive a Gold award.

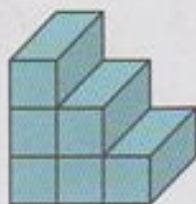
How many Bronze awards are needed to receive a Gold award?

- (A) 6×4
- (B) $6 + 4$
- (C) $6 \times 4 + 1$
- (D) $6 + 4 + 1$

21. Rachael made this shape from some small blocks.



How many small blocks did she take away to make this shape?



- (A) 12
- (B) 9
- (C) 6
- (D) 3

23. Tim stacked these boxes of biscuits for display in the supermarket.

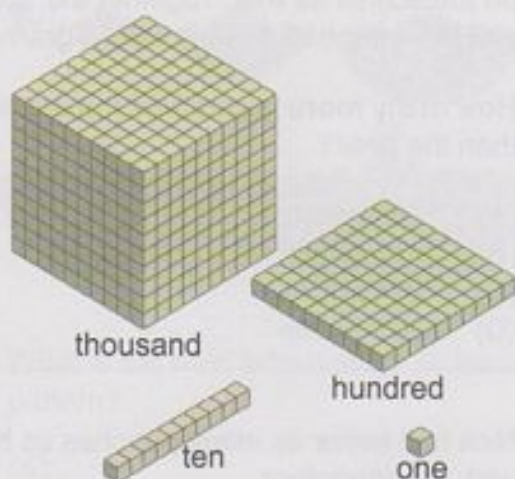
He made the display look the same on every side.



How many boxes did Tim use altogether in his display?

- (A) 8
- (B) 13
- (C) 14
- (D) 18

24. Sunil has blocks like these for showing numbers.



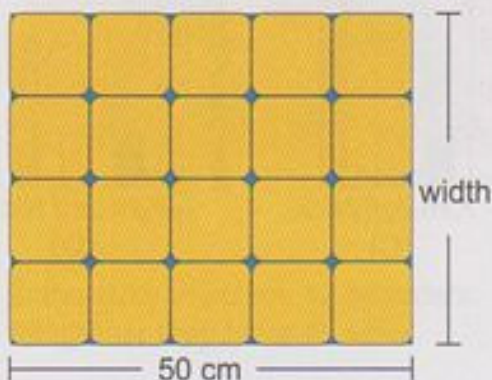
Sunil works out the answer to

$$77\,000 \div 9$$

How many thousand blocks does Sunil need to show his answer?

- (A) 6
- (B) 7
- (C) 8
- (D) 9

25.



What is the width of the rectangle?

- (A) 4 cm
- (B) 5 cm
- (C) 40 cm
- (D) 50 cm

26. Sue had this badge.



Which of these shows the same badge?



(A)



(B)



(C)



(D)

27. Lena saw this sign as she began her trip.

Pelican Point	35 km
Tern Village	105 km
Gulls Town	175 km

When she arrived at Tern Village, she drove back to Pelican Point. She then drove to Gulls Town.

What is the total distance Lena drove?

- (A) 210 km
- (B) 280 km
- (C) 315 km
- (D) 385 km

28.

Number	16	25	27
Multiple of 3	X	X	?
Multiple of 4	✓	X	?
Multiple of 5	X	✓	?
Multiple of 9	X	X	?

How many ticks should 27 have?

- (A) 1
(B) 2
(C) 3
(D) 4

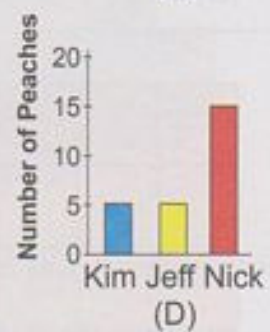
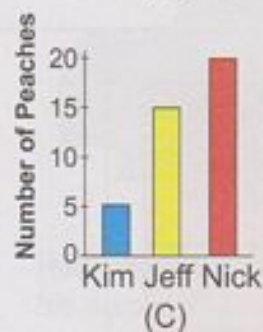
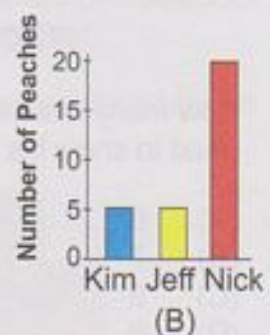
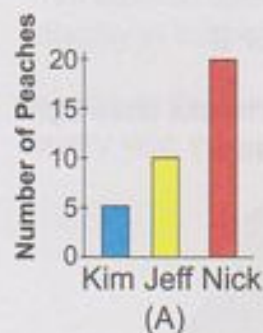
30. Jim put an apple on a set of scales. It had a mass of 316 grams. He then put a pear on the scales as well. Together the apple and the pear had a mass of 613 grams.

How many **more** grams was the apple than the pear?

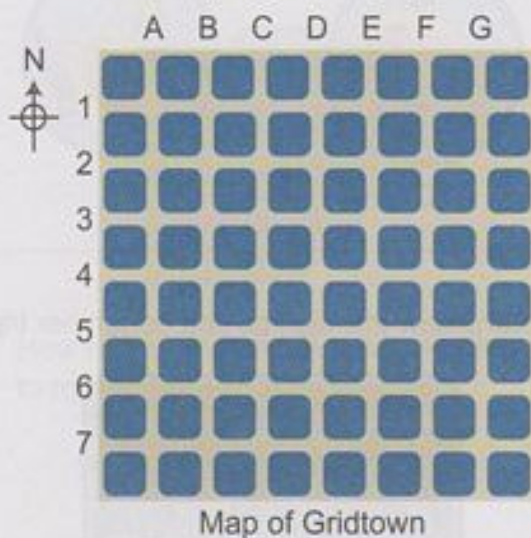
- (A) 13 grams
(B) 19 grams
(C) 297 grams
(D) 303 grams

31. Nick has **twice** as many peaches as Kim and Jeff altogether.

Which graph shows this information?



29. The picture shows a map of Gridtown.

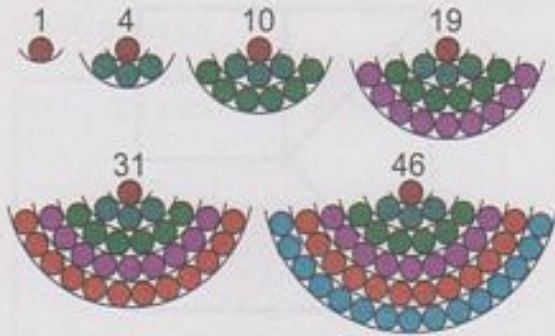


Jack is at A5 and Debbie is at E3. Phil is two streets north of Jack and two streets west of Debbie.

Where is Phil?

- (A) C3
(B) G7
(C) A3
(D) E1

32. Stefan made this number pattern.



What is the next **odd** number in Stefan's pattern?

- (A) 61
- (B) 64
- (C) 83
- (D) 85

33. Sally wrote a story about two kinds of monsters.

Sally's monsters



1 head
3 eyes

2 heads
4 eyes

In the story there are 12 monsters.
Altogether they have 46 eyes.

How many two-headed monsters are in the story?

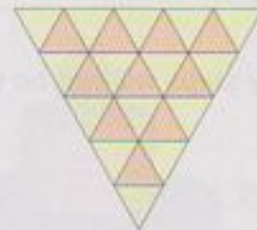
- (A) 2
- (B) 6
- (C) 10
- (D) 11

34. Alana, Mei and Kay were picking oranges. Alana picked 5 more oranges than Mei. Kay picked 8 more oranges than Mei.

Which of these is true?

- (A) Alana has 3 fewer oranges than Kay.
- (B) Kay has 3 fewer oranges than Alana.
- (C) Alana has 13 fewer oranges than Kay.
- (D) Kay has 13 fewer oranges than Alana.

35. This grid has a triangular pattern.



By cutting along the lines, triangles of different sizes can be made.

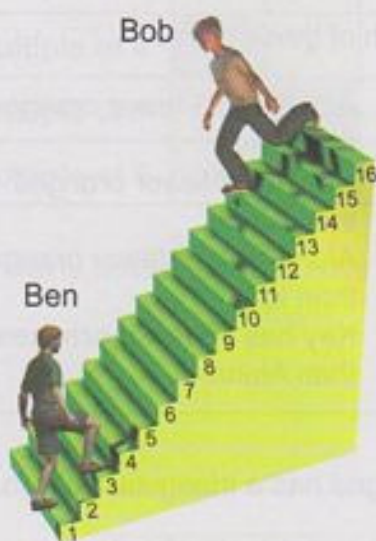


Sarab cuts the grid along the lines into triangles. At least two triangles are different in size. There are no pieces left over.

What is the smallest number of triangles Sarab can cut the grid into?

- (A) 4
- (B) 5
- (C) 8
- (D) 10

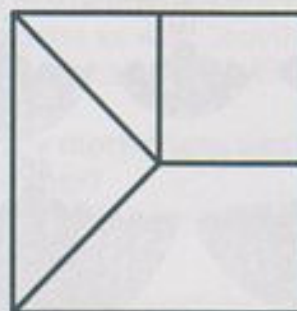
36. Bob walks down the stairs three steps at a time to step 1. Ben walks up the stairs two steps at a time to step 15.



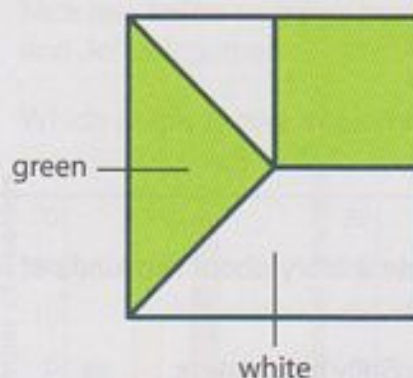
How many steps will **not** be walked on?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

37. This square has four small shapes.



Jacob colours two of the small shapes green. Here is one way he could do this.



In how many different ways can Jacob colour two of the small shapes green?

- (A) 3
- (B) 6
- (C) 8
- (D) 9

38. Bill is drawing people for his comic book. He drew some people with hats and some with glasses.

BILL'S COMIC BOOK PEOPLE		
	No glasses	Glasses
No hat		
Hat		

Bill counted the people in his comic book. There were 20 people wearing either a hat or glasses but not both. There were 6 people wearing both a hat and glasses. There were 11 people wearing hats.

How many people were wearing glasses, but **not** a hat?

- (A) 9
- (B) 14
- (C) 15
- (D) 16

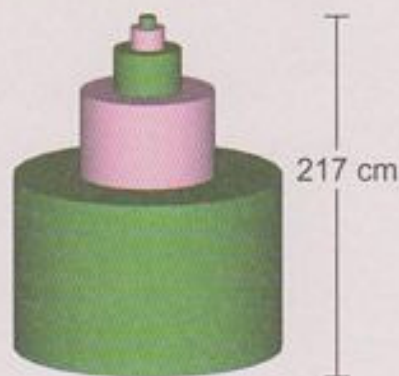
39. Jane, Ali and Pat each planted a tree 90 cm in height.

In the time it took Jane's tree to grow 1 cm, Ali's tree grew 2 cm. In the time it took Ali's tree to grow 1 cm, Pat's tree grew 3 cm.

How tall was Jane's tree when Pat's tree was 108 cm in height?

- (A) 18 cm
- (B) 36 cm
- (C) 93 cm
- (D) 96 cm

40. The picture shows a stack of five cylinders that is 217 cm high.



Each cylinder is half the height of the cylinder below it.

What is the height of the biggest cylinder?

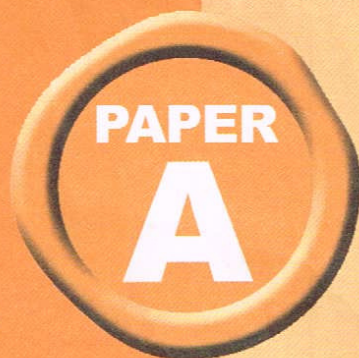
- (A) 43 cm
- (B) 56 cm
- (C) 108 cm
- (D) 112 cm

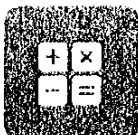
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SHOULD SIT THIS PAPER:**

AUSTRALIA: Year 3
BRUNEI: Primary 3
INDONESIA: Year 4
MALAYSIA: Standard 3
NEW ZEALAND: Year 4
PACIFIC: Year 3
SINGAPORE: Primary 2
SOUTH AFRICA: Grade 3





M

2009 Mathematics Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	C	A	B	A	D	C	A	C
2	C	B	D	D	C	B	A	B
3	A	D	B	C	A	D	C	C
4	B	D	D	D	C	C	C	C
5	D	A	D	D	B	D	D	D
6	A	D	A	B	A	A	D	C
7	B	C	A	C	C	A	B	A
8	C	B	D	C	B	C	B	A
9	D	A	C	B	A	B	A	D
10	A	D	C	C	D	C	C	C
11	D	B	B	A	D	D	B	B
12	A	B	D	D	B	B	C	D
13	B	C	B	B	B	C	B	C
14	D	B	D	A	C	B	A	D
15	A	D	C	B	C	B	C	B
16	D	B	C	A	B	A	C	B
17	B	A	D	A	D	C	B	D
18	B	A	B	D	A	B	D	A
19	D	B	A	A	A	D	B	D
20	D	C	B	B	B	C	D	C
21	A	D	A	B	D	A	A	A
22	A	D	A	C	A	A	B	A
23	D	C	D	C	B	D	D	C
24	C	A	A	B	B	B	D	B
25	C	C	B	D	B	B	A	B
26	A	D	B/D	B	C	D	D	D
27	C	C	A	D	D	A	D	D
28	B	A	B	A	A	C	B	C
29	A	B	C	B	C	B	C	A
30	B	C	C	C	B	D	D	A
31	B	A	A	C	D	A	D	B
32	D	B/D	C	A	C	A	B	D
33	C	A	A	C	B	D	A	A
34	A	C	C	D	D	A	A	B
35	C	C	C	B	A	D	C	C
36	D	D	B	18 018	132	116	116	528
37	B	B	D	84 084	18 018	256	324	24 024
38	C	B	B	390	400	24 024	52 052	10 010
39	C	C	C	10 010	24 024	2 02 002	528	76 076
40	D	D	C	361	21 021	970	484	484

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